UGANDA CHRISTIAN UNIVERSITY
A Centre of Excellence in the Heart of Africa

FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY
DEPARTMENT OF COMPUTING AND TECHNOLOGY
ADVENT 2026 SEMESTER TEST ONE

PROGRAM: BACHELOR OF SCIENCE IN COMPUTER SCIENCE
BACHELOR OF SCIENCE IN DATA SCIENCE

YEAR: 1

SEMESTER: 1

COURSE CODE: MTH1203

COURSE NAME: PROBABILITY AND STATISTICS

EXAMINATION TYPE: PRACTICAL -100%

EXAMINATION DATE: SEPTEMBER 2026

TIME ALLOWED: 1 HOURS

Examination Instructions

1. The general Uganda Christian University examination guidelines and academic & financial policies apply to this examination. Violating any of the policies by the student automatically makes this examination attempt void, even if you have completed and submitted the answer booklet.
2. Attempt all the questions. Every question carries 20 marks.
3. The examination should be executed in Python, and answers to every question should be placed in a new Jupyter Notebook.
4. Every question must include written interpretation: state and explain each result in a Markdown cell, not in code comments alone.
5. Submit a zipped folder containing Jupyter Notebooks and any other exported files.
6. Uganda Christian University or its officials reserve the right of admission or disqualification from this exam.

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Question 1

The table below shows the end-of-semester record for 12 first-year students. This data is **not** in a file; you must create the DataFrame yourself.

Student	Programme	Sex	StudyHours	Attendance	Score	Result	Scholarship
Aisha	BSDS	F	18	92	78	Pass	Yes
Brian	BSCS	M	8	60	45	Fail	No
Cathy	BSCS	F	22	95	88	Pass	Yes
Derek	BSIT	M	6	55	39	Fail	No
Esther	BSDS	F	20	90	83	Pass	Yes
Felix	BSCS	M	12	70	61	Pass	No
Gloria	BSIT	F	14	80	66	Pass	No
James	BSDS	M	10	65	52	Pass	No
Annet	BSDS	F	24	98	91	Pass	Yes
Daniel	BSCS	M	7	58	44	Fail	No
Brenda	BSIT	F	9	62	48	Fail	No
Oscar	BSDS	M	16	85	72	Pass	Yes

Table 1: End-of-semester record for 12 first-year students.

(a) **4 marks**

Create a pandas DataFrame called `results_df` from the table above. State the **expected** number of rows and columns in a Markdown cell **before** running the code, then verify with `.shape` and `.info()`, confirm there are no missing values, and note which columns are categorical.

(b) **4 marks**

Compute the mean, median and standard deviation of `StudyHours`, `Attendance` and `Score`. Identify the highest- and lowest-scoring students, and comment on how spread out the scores are.

(c) **4 marks**

For the `Score` column, compute the quartiles `Q1`, `Q2` (the median) and `Q3`, the interquartile range (IQR) and the 90th percentile. Using the $1.5 \times$ IQR rule, report the outlier boundaries and state whether any student is an outlier on `Score`. Then use `.describe()` and compare the mean and median of `Score` to comment on whether the distribution is symmetric or skewed.

(d) **4 marks**

Split the students into those who passed and those who failed (the `Result` column). Compute the mean `StudyHours` and mean `Attendance` for each group, and state which of the two factors shows the larger gap between the pass and fail groups.

(e) **4 marks**

Synthesise your findings from (b)–(d) into a data-driven recommendation (120–150 words) to the scholarship committee on which single factor best identifies strong students. State one limitation of drawing conclusions from only 12 records.

Question 2

The `penguins.csv` dataset (Palmer Archipelago) contains field measurements of 344 penguins. Each record includes the following variables:

Column	Type	Description
`species`	Categorical	Penguin species (Adelie / Chinstrap / Gentoo)
`island`	Categorical	Island of observation (Biscoe / Dream / Torgersen)
`bill_length_mm`	Numeric	Length of the bill (mm)
`bill_depth_mm`	Numeric	Depth of the bill (mm)
`flipper_length_mm`	Numeric	Flipper length (mm)
`body_mass_g`	Numeric	Body mass (g)
`sex`	Categorical	Sex of the penguin (male / female)

A researcher wishes to summarise the physical profile of the penguins and draw conclusions about the wider population from which they were sampled.

(a) **3 marks**

(i) Load the file into a DataFrame `penguins`, then create a working copy `data` that drops all rows containing missing values.

(ii) Display the first 5 rows and report the shape (rows, columns) **before and after** dropping missing values. State how many rows were lost.

(b) **5 marks**

(i) For `bill_length_mm`, `flipper_length_mm` and `body_mass_g`, compute the mean, median, mode, standard deviation and interquartile range (IQR). Present the results in a single summary table.

(ii) Classify each column of `data` as **nominal**, **ordinal**, **discrete** or **continuous**, justifying each choice in one line.

(c) **4 marks**

(i) Compute the mean `body_mass_g` for each `species`. State which species is heaviest and which is lightest.

(ii) For each `species`, compute the interquartile range (IQR) of `body_mass_g`, and use the $1.5 \times$ IQR rule to report how many penguins (if any) are outliers within each species. State which species has the largest spread.

(d) **4 marks**

For `body_mass_g`, compute the 10th, 25th, 50th (median), 75th and 90th percentiles. Interpret, in plain words, what the 90th percentile tells you about the penguins, and report how many penguins weigh **more** than the 90th-percentile value.

(e) **4 marks**

For `flipper_length_mm`, compute the mean, standard deviation and range for each `species`, and present the results as a single grouped summary table. State which species has the longest flippers on average, and which species is the most variable (largest standard deviation).

◀ END ▶

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